

Spotify 2023 Data Analysis using Power BI

Brief

You have been provided with the **Spotify 2023 dataset** ([spotify-2023.csv](#)), containing information about popular songs, artists, streaming activity, playlist presence, chart performance, and musical characteristics.

For this coursework, consider yourself a **Business Intelligence Analyst in the music industry**. Using **Microsoft Power BI**, analyse the dataset to identify meaningful relationships, patterns, differences, and exceptional cases that can provide useful insights for music-industry decision-making.

You should prepare the dataset where necessary and develop an **interactive Power BI analytical report**. Your analysis should focus on explaining what the available data reveals rather than predicting future performance.

Analysis Tasks

1. Platform Comparison

Compare the popularity and performance of songs across different music platforms.

Analyse available indicators for **Spotify, Apple Music, Deezer, and Shazam**, including streams, playlists, and chart presence.

Identify differences in platform performance and examine whether songs that perform strongly on one platform also demonstrate strong performance on other platforms.

2. Artist Impact

Analyse how **artist involvement and artist-related attributes** are associated with song performance.

Investigate variables such as **number of artists involved, streams, playlist presence, chart performance, and other available song characteristics**.

Compare songs involving different numbers of artists and identify meaningful patterns in their performance.

3. Playlist Exposure and Chart Performance

Investigate the relationship between **playlist presence and chart performance**.

Analyse whether songs with greater playlist exposure also demonstrate stronger positions or presence in music charts.

Compare playlist exposure with available **Spotify, Apple Music, and Deezer chart indicators**, and use Power BI visualizations to identify important relationships and differences.

4. Shazam Discovery Analysis

Investigate the relationship between **Shazam activity and song performance**.

Analyse Shazam chart presence alongside Spotify streams, playlist presence, and other available platform indicators.

Identify songs that have strong Shazam presence and examine whether they also perform strongly on other platforms. Highlight any notable differences or exceptions in the data.

5. Major vs Minor: Key and Mode Analysis

Investigate whether songs in different **musical keys and modes** demonstrate different patterns in streaming performance and audio characteristics.

Compare **major and minor mode songs** using variables such as streams, danceability, energy, valence, acousticness, and BPM.

Use appropriate Power BI visualizations to identify differences between the groups and discuss what the data indicates about musical characteristics and performance.

6. BPM and Song Performance Analysis

Analyse the relationship between **tempo (BPM)** and song performance.

Group songs into appropriate BPM ranges and compare their **streams, audio characteristics, playlist presence, and chart performance**.

Identify whether particular tempo ranges are more common among highly streamed songs and investigate whether BPM is associated with other musical characteristics in the dataset.

Instructions

1. The analysis must be completed using **Microsoft Power BI**.
2. Students should import and prepare the supplied `spotify-2023.csv` dataset before beginning the analysis.
3. Appropriate data transformation and cleaning should be performed where necessary.
4. Students may create **calculated columns and DAX measures** where appropriate.
5. Each analysis should contain suitable Power BI visualizations and should be accompanied by an explanation of the findings.
6. Students should use appropriate **KPIs, slicers, filters, charts, tables, drill-downs, and interactive features** to improve the analytical report.
7. Visualizations should be selected based on the analytical question rather than simply displaying as many charts as possible.
8. Students must justify the analytical approach and visualization choices used.
9. Findings should be supported by evidence from the dataset.
10. Any assumptions, data limitations, or potential data-quality issues should be clearly documented.
11. Students should distinguish between **observed relationships and causal claims**.
12. **Forecasting, prediction, and machine-learning models are not required.**
13. The final Power BI report should provide clear, concise, and actionable insights relevant to the music industry.